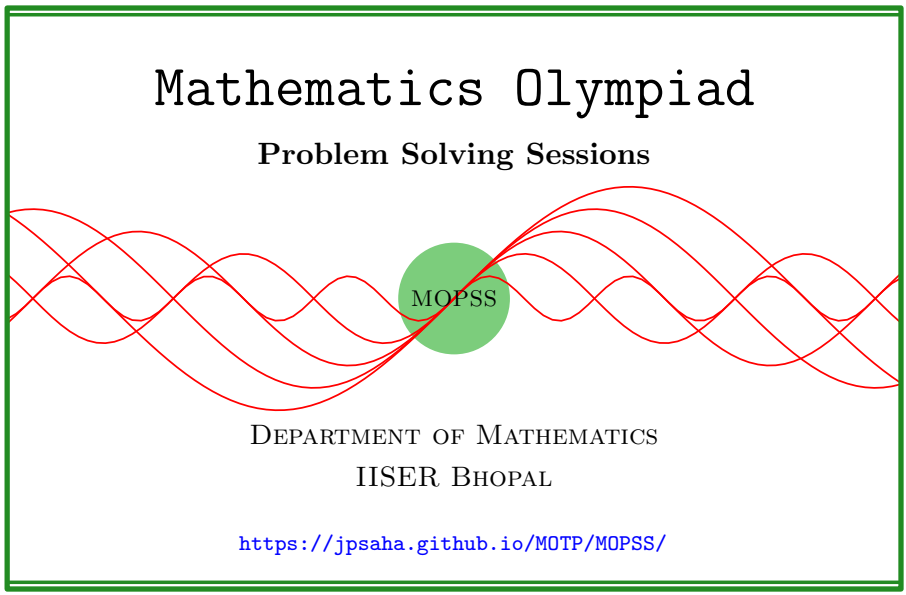
Roots of unity

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

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3 Exercise (USAMO 1976 P5,) 4
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§1 Roots of unity

Some of the problems have been borrowed from Yufei Zhao's handout on polynomials.

Exercise 1 (Pre-RMO 2012 P17, cf. Exercise 2, USAMO 2014 P1)**.** Let x_1, x_2, x_3 be the roots of the equation $x^3 + 3x + 5 = 0$. What is the value of the expression

$$\left(x_1 + \frac{1}{x_1}\right)\left(x_2 + \frac{1}{x_2}\right)\left(x_3 + \frac{1}{x_3}\right)?$$

Solution 1. Let $P(x)$ denote the polynomial $x^3 + 3x + 5$. Note that

$$\begin{aligned}
&\left(x_1 + \frac{1}{x_1}\right)\left(x_2 + \frac{1}{x_2}\right)\left(x_3 + \frac{1}{x_3}\right) \\
&= \frac{1}{x_1x_2x_3}(x_1^2+1)(x_2^2+1)(x_3^2+1) \\
&= \frac{1}{x_1x_2x_3}(x_1+i)(x_2+i)(x_3+i)(x_1-i)(x_2-i)(x_3-i) \\
&= \frac{1}{x_1x_2x_3}P(-i)P(i) \\
&= \frac{1}{-5}\left|P(i)\right|^2 \\
&= \frac{1}{-5}\left|5-2i\right|^2 \\
&= -\frac{29}{5}.
\end{aligned}$$

■

Remark. The above argument is elegant and quite useful. One could have also

The content posted here and at this blog by Evan Chen are quite useful.

argued that

$$\begin{aligned}
&\left(x_1+\frac{1}{x_1}\right)\left(x_2+\frac{1}{x_2}\right)\left(x_3+\frac{1}{x_3}\right)\\
&=\left(x_1x_2+\frac{x_1}{x_2}+\frac{x_2}{x_1}+\frac{1}{x_1x_2}\right)\left(x_3+\frac{1}{x_3}\right)\\
&=x_1x_2x_3+\frac{x_1x_3}{x_2}+\frac{x_2x_3}{x_1}+\frac{x_3}{x_1x_2}+\frac{x_1x_2}{x_3}+\frac{x_1}{x_2x_3}+\frac{x_2}{x_3x_1}+\frac{1}{x_1x_2x_3}\\
&=x_1x_2x_3++\frac{1}{x_1x_2x_3}+\frac{x_1x_2}{x_3}+\frac{x_2x_3}{x_1}+\frac{x_3x_1}{x_2}+\frac{x_1}{x_2x_3}+\frac{x_2}{x_3x_1}+\frac{x_3}{x_1x_2}\\
&=-5-\frac{1}{5}+\frac{1}{x_1x_2x_3}\left(x_1^2x_2^2+x_2^2x_3^3+x_3^2x_1^2\right)+\frac{1}{x_1x_2x_3}\left(x_1^2+x_2^2+x_3^2\right)\\
&=-5-\frac{1}{5}-\frac{1}{5}\left((x_1x_2+x_2x_3+x_3x_1)^2-2x_1x_2x_3(x_1+x_2+x_3)\right)\\
&-\frac{1}{5}\left((x_1+x_2+x_3)^2-2(x_1x_2+x_2x_3+x_3x_1)\right)\\
&=-5-\frac{1}{5}-\frac{1}{5}\left(3^2\right)-\frac{1}{5}\left(-2\cdot 3\right)\\
&=-5-\frac{1}{5}-\frac{9}{5}+\frac{6}{5}\\
&=-5-\frac{4}{5}\\
&=-\frac{29}{5}.
\end{aligned}$$

Note that this way of arguing would get complication if we had a higher degree polynomial to start with.

Exercise 2 (USAMO 2014 P1,). Let a, b, c, d be real numbers such that $b-d \geq 5$ and all zeros x_1, x_2, x_3, x_4 of the polynomial $P(x)=x^4+ax^3+bx^2+cx+d$ are real. Find the smallest value the product

$$(x_1^2+1)(x_2^2+1)(x_3^2+1)(x_4^2+1)$$

can take.

See also Exercise 1, India Pre-RMO 2012 P17.

Walkthrough —

(a)

Solution 2. Note that

$$\begin{aligned}
(x_1^2+1)(x_2^2+1)(x_3^2+1)(x_4^2+1) &= P(i)P(-i)\\
&=(1-b+d)^2+(a-c)^2\\
&=(b-d-1)^2+(a-c)^2
\end{aligned}$$

$$\geq 16.$$

Taking $a = c$ and $b = 5, d = 0$, we obtain

$$(x_1^2+1)(x_2^2+1)(x_3^2+1)(x_4^2+1) = 16.$$

Hence, the smallest value the product $(x_1^2+1)(x_2^2+1)(x_3^2+1)(x_4^2+1)$ can take is equal to 16. ■

Exercise 3 (USAMO 1976 P5,). If $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ are all polynomials such that

$$P(x^5) + xQ(x^5) + x^2R(x^5) = (x^4+x^3+x^2+x+1)S(x),$$

prove that $x-1$ is a factor of $P(x)$.

Summary — The primitive 5-th roots of unity can be used to show that $P(1) = 0$.

Walkthrough —

(a) Substituting primitive 5-th roots of unity (that is, the 5-th roots of unity other than 1) for x, yields several linear equations in $P(1), Q(1), R(1)$.

(b) Can $Q(1)$ and $R(1)$ be eliminated to obtain that $P(1) = 0$?

Solution 3. Denote the 5-th root of unity $\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$ by ζ. Substituting $x = \zeta, \zeta^2, \zeta^3$, we obtain

$$\begin{aligned} P(1) + \zeta Q(1) + \zeta^2 R(1) &= 0, \\ P(1) + \zeta^2 Q(1) + \zeta^4 R(1) &= 0, \\ P(1) + \zeta^3 Q(1) + \zeta^6 R(1) &= 0. \end{aligned}$$

Eliminating $R(1)$ from the first two equations yields

$$(1-\zeta^2)P(1) + \zeta^2(1-\zeta)Q(1) = 0,$$

and eliminating $R(1)$ from the last two equations yields

$$(1-\zeta^2)P(1) + \zeta^3(1-\zeta)Q(1) = 0.$$

Eliminating $Q(1)$ from the above two equations, we obtain $(1-\zeta)P(1) = 0$, which gives $P(1) = 0$. This shows that $x-1$ is a factor of $P(x)$. ■

Exercise 4 (Leningrad Math Olympiad 1991 P6)**.** A finite sequence $a_1, a_2, \ldots, a_n$ is called p-balanced if any sum of the form

$$a_k + a_{k+p} + a_{k+2p} + \ldots$$

is the same for any $k = 1, 2, 3, \ldots, p$. For instance the sequence

$$a_1 = 1, a_2 = 2, a_3 = 3, a_4 = 4, a_5 = 3, a_6 = 2$$

is a 3-balanced. Prove that if a sequence with 50 members is p-balanced for $p = 3, 5, 7, 11, 13, 17$, then all its members are equal zero.

Summary — Consider the polynomial $\sum_{i=1}^{50} a_i x^i$.

Walkthrough —

(a) Show that the polynomial vanishes at any p-th root of unity, other than 1, for $p \in \{3, 5, 7, 11, 13, 17\}$.

(b) How many such roots of unity are there in total?

Solution 4. Let $a_1, a_2, \ldots, a_{50}$ be a sequence of complex numbers. Assume that it is p-balanced for $p \in \{3, 5, 7, 11, 13, 17\}$. For an integer $n \geq 1$, denote the root of unity $\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$ by ζ_n. Let $P(x)$ denote the polynomial $\sum_{i=1}^{n} a_i x^i$.

Let $3 \leq p \leq 17$ be a prime. Since $a_1, a_2, \ldots, a_{50}$ is p-balanced, for any $1 \leq \ell < p$, we obtain

$$\begin{aligned} P(\zeta_p^\ell) &= \sum_{k=1}^{p} (a_k + a_{k+p} + \ldots) \zeta_p^{k\ell} \\ &= (a_1 + a_{1+p} + \ldots) \sum_{k=1}^{p} \zeta_p^{k\ell} \\ &= 0, \end{aligned}$$

where the final equality follows since $\zeta_p^\ell \neq 1$. This shows that the polynomial $P(x)$ vanishes at the elements of the set

$$\cup_{p \in \{3,5,7,11,13,17\}} \{\zeta_p^\ell \mid 1 \leq \ell < p\},$$

which contains

$$\sum_{p \in \{3,5,7,11,13,17\}} (p-1) = 2 + 4 + 6 + 10 + 12 + 16 = 50$$

elements. Moreover, $P(x)$ also vanishes at 0. Note that $P(x)$ is a polynomial of degree 50, and it has at least 51 zeroes. This gives that $P(x) = 0$, and hence, the terms of the sequence $a_1, a_2, \ldots, a_{50}$ are all equal to zero. ■

Exercise 5 (USAMO 1977 P1,). Determine all pairs of positive integers (m, n) such that $1 + x^n + x^{2n} + \cdots + x^{mn}$ is divisible by $1 + x + x^2 + \cdots + x^m$.

Solution 5. Let μ_{m+1} denote the set of all $(m+1)$-th roots of unity in $\mathbb{C}$. Note that

$$1 + x + x^2 + \cdots + x^m = \frac{x^{m+1} - 1}{x - 1} = \prod_{\zeta \in \mu_{m+1} \setminus \{1\}} (x - \zeta).$$

Moreover, we have

$$1 + x^n + x^{2n} + \cdots + x^{mn} = \prod_{\xi \in \mu_{m+1} \setminus \{1\}} (x^n - \xi).$$

If $m+1, n$ are relatively prime, then the map $x \mapsto x^n$ is a bijection on μ_{m+1}, and it fixes the point 1, and hence,

$$1 + x^n + x^{2n} + \cdots + x^{mn} = \prod_{\zeta \in \mu_{m+1} \setminus \{1\}} (x^n - \zeta^n),$$

which shows that $1 + x + x^2 + \cdots + x^m$ divides $1 + x^n + x^{2n} + \cdots + x^{mn}$.

Let d denote the greatest common divisor of $m+1$ and n. Assume that $d > 1$. Then, there exists a $(m+1)$-th root of unity ζ such that $\zeta^n = 1$ and $\zeta \neq 1$. Indeed, let ω be a primitive $(m+1)$-th root of unity. Then, $\omega^{(m+1)/d}$ is such a root of unity. Note that

$$1 + x^n + x^{2n} + \cdots + x^{mn}$$

does not vanish at $x = \zeta$, and hence $1 + x + x^2 + \cdots + x^m$ does not divide $1 + x^n + x^{2n} + \cdots + x^{mn}$. This shows that $1 + x + x^2 + \cdots + x^m$ divides $1 + x^n + x^{2n} + \cdots + x^{mn}$ if and only if $m+1, n$ are relatively prime. ■

Example 6. Show that any function $f : \mathbb{R} \to \mathbb{R}$ can be written as a sum of an even function and an odd function.

Solution 6. Consider the functions $f_1, f_{-1} : \mathbb{R} \to \mathbb{R}$ defined by

$$f_\varepsilon(x) = \frac{1}{2}(f(x) + \varepsilon f(\varepsilon x)), \qquad \text{for all } x \in \mathbb{R},$$

where $\varepsilon \in \{1, -1\}$. Note that f_1 is an even function, f_{-1} is an odd function, and that $f = f_1 + f_{-1}$ holds. ■

> **Remark.** It seems from the above solution that we "knew" the functions f_1, f_{-1} beforehand! Here is another solution to the above problem.
>
> Let us assume that f can be expressed as the sum of an even function g and an odd function h. Note that
>
> $$f(x) = g(x) + h(x),$$

$$\begin{aligned} f(-x) &= g(-x) + h(-x) \\ &= g(x) - h(x) \end{aligned}$$

hold for all $x \in \mathbb{R}$. Combining the above, we obtain

$$g(x) = \frac{1}{2}(f(x) + f(-x)),$$
$$h(x) = \frac{1}{2}(f(x) - f(-x))$$

for all $x \in \mathbb{R}$.

Note that this is **NOT a proof**. We have observed that

$$g(x) = \frac{1}{2}(f(x) + f(-x)),$$
$$h(x) = \frac{1}{2}(f(x) - f(-x))$$

holds **relying on the assumption that** g is an even function, and h is an odd function satisfying $f = g + h$. This observation is **made under the hypothesis** that $f = g + h$ where g (resp. h) is even (resp. odd), and **cannot be a proof of the hypothesis**! However, this observation can be used to arrive at the above proof, by defining the functions f_1, f_{-1} as above, and showing that they have the required properties.

Example 7. Let ω denote the root of unity $\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$. Show that any function $f : \mathbb{C} \to \mathbb{C}$ can be written as a sum of three functions $f_0, f_1, f_2 : \mathbb{C} \to \mathbb{C}$ satisfying

$$f_i(\omega z) = \overline{\omega}^i z \quad \text{for all } z \in \mathbb{C}.$$

Walkthrough — Assume that the function f can be expressed as stated, and then determine some of the properties of the functions f_0, f_1, f_2, as done in the above remark. What remains left to do?

Example 8. Let $P(x)$ be a monic polynomial with integer coefficients such that all its zeroes lie on the unit circle. Show that all the zeroes of $P(x)$ are roots of unity, that is, $P(x)$ divides $(x^n - 1)^k$ for some positive integers n, k.

The author has learnt the following argument from Mainak Ghosh, during the INMO Training Camp 2025, held at IISER Bhopal.

Walkthrough —

(a) Use the fundamental theorem of symmetric polynomials, to prove the following claim.

> **Claim** — Let $f(x)$ be a monic polynomial of degree n with integer coefficients. Let $\alpha_1, \ldots, \alpha_n$ denote its roots, counting multiplicities. Then for any integer $k \geq 1$, there is a monic polynomial of degree n with integer coefficients, having $\alpha_1^k, \alpha_2^k, \ldots, \alpha_n^k$ as its roots.

(b) Applying the Claim, for each integer $k \geq 1$, obtain a monic polynomial $P_k(x)$ with integer coefficients, having degree same as that of $P(x)$, whose roots, counted with multiplicities, are the k-th powers of the roots of $P(x)$.

(c) Note that the polynomials $P_1(x), P_2(x), \ldots$ are of the same degree, and the absolute values of the coefficients of any of them are bounded from the above by suitable binomial coefficients, which is smaller than 2^n, where n denotes the degree of $P(x)$. Since these polynomials have integer coefficients, by the pigeonhole principle, it follows that there is a positive integer k_0 such that $P_k(x) = P_{k_0}(x)$ holds for infinitely many positive integers k.

(d) Enumerate the roots of $P(x)$, and matching the roots of $P_k(x)$ with those of $P_{k_0}(x)$ (for suitable k's), we obtain a permutation of n letters. By the pigeonhole principle, infinitely many k's yield the same permutation, which implies that there are positive integers $k \neq \ell$ such that for any root of $P(x)$, its k-th and the ℓ-th powers are equal.

Solution 7. ■

Exercise 9 (Harvard-MIT Mathematics Tournament 2023 February, Algebra and Number Theory P10, , proposed by Sean Li)**.** Let $\zeta = e^{2\pi i/99}$ and $\omega = e^{2\pi i/101}$. The polynomial

$$x^{9999} + a_{9998}x^{9998} + \cdots + a_1 x + a_0$$

has roots $\zeta^m + \omega^n$ for all pairs of integers (m, n) with $0 \leq m < 99$ and $0 \leq n < 101$. Compute $a_{9799} + a_{9800} + \cdots + a_{9998}$.

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)